

Genome-wide identification and characterization of aquaporin gene family in common bean (*Phaseolus vulgaris* L.)

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Abstract Plant aquaporins are a large and diverse family of water channel proteins that are essential for several physiological processes in living organisms. Numerous studies have linked plant aquaporins with a plethora of processes, such as nutrient acquisition, CO₂ transport, plant growth and development, and response to abiotic stresses. However, little is known about this protein family in common bean. Here, we present a genome-wide identification of the aquaporin gene family in common bean (*Phaseolus vulgaris* L.), a legume crop essential for human nutrition. We identified 41 full-length coding aquaporin sequences in the common bean genome, divided by phylogenetic analysis into five sub-families (PIPs, TIPs, NIPs, SIPs and XIPs). Residues determining substrate specificity of aquaporins (i.e., NPA motifs and ar/R selectivity filter) seem conserved between common bean and other plant species, allowing inference of substrate specificity for these proteins. Thanks to the availability of RNA-sequencing datasets, expression levels in different organs and in leaves of wild and domesticated bean accessions were evaluated. Three aquaporins (PvTIP1;1, PvPIP2;4 and PvPIP1;2) have the overall highest mean expressions, with PvTIP1;1 having the highest expression among all aquaporins. We performed an EST database mining to identify drought-responsive

aquaporins in common bean. This analysis showed a significant increase in expression for PvTIP1;1 in drought stress conditions compared to well-watered environments. The pivotal role suggested for PvTIP1;1 in regulating water homeostasis and drought stress response in the common bean should be verified by further field experimentation under drought stress.

Keywords Aquaporin · Drought response · Gene expression · Gene structure · Water homeostasis

Introduction

Aquaporins are water channel proteins that facilitate diffusion of water and small molecules across biological membranes (Gomes et al. 2009). In plants, aquaporins showed surprisingly diverse and numerous isoforms per species, with more than 30 members identified in different plant genomes (Chaumont et al. 2001; Johanson et al. 2001; Reuscher et al. 2013). Analysis of the *Arabidopsis* genome, the first plant with a completely sequenced genome (AGI 2000), identified four sub-families of aquaporins, namely plasma membrane intrinsic proteins (PIPs), with two groups (PIPs1 and PIPs2); tonoplast intrinsic proteins (TIPs); nodulin 26-like intrinsic proteins (NIPs) and small intrinsic proteins (SIPs) (Johanson et al. 2001; Gomes et al. 2009). The sequencing of other plant genomes also identified an additional sub-family absent in *Arabidopsis*, named X-intrinsic proteins (XIPs) (Gupta and Sankararamakrishnan 2009; Bienert et al. 2011; Reuscher et al. 2013).

Aquaporins assemble in biological membranes as homo- or hetero-tetramers in which each monomer forms a water channel (Chaumont et al. 2005). Different experiments have shown that tetramerization between members

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of the PIP1 and PIP2 sub-families modifies both water channel activity and protein sub-cellular localization (Fetter et al. 2004; Zelazny et al. 2007, 2009). The main features of aquaporin protein monomers are highly conserved in all living organisms, consisting of six transmembrane domains with a central aqueous pore characterized by two NPA motifs (Asp-Pro-Ala). These motifs are critical for the transport function of these proteins (Murata et al. 2000). Other important residues in aquaporin sequences are the ones forming the aromatic/arginine selectivity filter (ar/R). This region is characterized by four residues, one in helix 2 (H2), one in helix 5 (H5), and two in loop E (LE1 and LE2). The ar/R selectivity filter acts as a size-exclusion barrier and regulates the transport specificity of these proteins (Hub and de Groot 2008; Mitani-Ueno et al. 2011).

Aquaporins are involved in almost every physiological process in plants, such as cell differentiation and elongation, plant transpiration, and regulation of plant hydraulics (Maurel 2007; Maurel et al. 2008; Li et al. 2014). Experimental evidence points to variations in expression or in protein accumulation in response to environmental stresses, hormone treatments, and the ability to transport small molecules like ammonia, urea, boron, silicon and CO₂ (Maurel et al. 2008; Li et al. 2014). Several experiments identified aquaporins as involved in response to drought stress in common beans varieties, and marked differences in gene expression were observed in drought-resistant versus susceptible genotypes (Aroca et al. 2006; Montalvo-Hernández et al. 2008; Recchia et al. 2013). In particular, a tolerant genotype showed increased expression of a tonoplast aquaporin in drought-stressed roots, with re-localization of its mRNA in phloem tissues (Montalvo-Hernández et al. 2008).

Because it is a major source of protein, complex carbohydrates, and micronutrients, the common bean (*P. vulgaris* L.) is one of the most important crops for human consumption worldwide (Broughton et al. 2003). Wild common bean is organized into two genetically different gene pools (Mesoamerican and Andean) (Koenig and Gepts 1989; Kwak and Gepts 2009; Mamidi et al. 2013), with a broad environmental distribution in Mexico and Central to South America (Cortés et al. 2013). These two gene pools were independently domesticated in what is now Mexico and South America (Gepts et al. 1986; Bitocchi et al. 2013) about 8,000 years ago. The release of a reference *P. vulgaris* genome sequence (Schmutz et al. 2014), together with the development and application of Next Generation Sequencing technologies in plant species (Bräutigam and Gowik 2010) and synteny among plants, such as soybean and *Arabidopsis*, enabled a genome-wide identification of actual or putative loci involved in the domestication process of this species (Repinski et al. 2012; Kwak et al. 2012; Bellucci et al. 2014; Schmutz et al. 2014).

The availability of a reference genome sequence also allows the identification and characterization of important plant gene families, thus facilitating genomic, evolutionary, and molecular studies in different species. From the current genome sequence of *P. vulgaris*, we identified the full set of aquaporin-coding sequences in the bean genome and characterized them at phylogenetic and structural levels. We also identified single nucleotide polymorphism (SNP) markers from the BARCBEAN6K_3 array (Beancap; www.beancap.org) located near the identified aquaporins. We then analyzed the expression of these genes in different plant organs and in different wild and domesticated bean accessions using publicly available expression datasets (see “Materials and methods”). Furthermore, we integrated the previously characterized aquaporins in a proposed nomenclature. We then analyzed the expression of aquaporin genes in response to drought stress using ESTs resources available at NCBI. Overall, this first genome-wide identification and analysis of common bean aquaporins provide a foundation for identifying and further functionally characterizing important aquaporins in common bean. Eventually, this information can lead to molecular breeding improvement of drought-related traits in this species.

Materials and methods

Identification of aquaporin genes and phylogenetic analysis

All the sequence-related information (gene sequences, reference genome, protein sequences, annotation and features) was retrieved from Phytozome (<http://phytozome.jgi.doe.gov/pz/portal.html>) (Goodstein et al. 2012). The multiple sequence alignments and phylogenetic analyses were performed using the Seaview toolkit (Guoy et al. 2010). For multiple sequence alignments, the MUSCLE algorithm, with default parameters, was used. Pairwise sequence alignments were performed on protein sequence with the needle algorithm of the EMBOSS suite using default parameters (Rice et al. 2000). *P. vulgaris* aquaporins were identified using a Hidden Markov Models (HMMs) probabilistic method implemented in HMMER3 (<http://hmmer.janelia.org/>). In brief, the complete set of aquaporin protein sequences of *Arabidopsis*, identified by Johanson et al. (2001), was aligned with the MUSCLE algorithm for building a HMM profile. This profile was queried to the full proteome of *P. vulgaris*. The resulting putative aquaporins were aligned with MUSCLE and manually checked for the presence of the two motifs forming the aqueous pore typical of aquaporins. Proteins without these two motifs were discarded from the analysis.

For phylogenetic analysis and nomenclature, *Arabidopsis* and *Populus trichocarpa* (Johanson et al. 2001; Lopez et al. 2012) aquaporin protein sequences, with their proposed nomenclature, were aligned with the *P. vulgaris* aquaporins using MUSCLE. A Bio Neighbor-Joining (BioNJ) tree, based on observed distance with 1000 bootstrap replicates, was built.

Tandem duplicated genes were identified as those with adjacent chromosomal localization, as determined also by the ID of the loci.

Gene structure, conserved residues, and in silico prediction of transmembrane domains and sub-cellular localization

Gene structure for each aquaporin was illustrated with the Gene Structure Display Server (<http://gsds.cbi.pku.edu.cn/>). Conserved residues of the NPA domain and ar/R selectivity filter (Murata et al. 2000; Hub and de Groot 2008; Mitani-Ueno et al. 2011) were identified by multiple sequence alignment with MUSCLE of the different sub-families of *P. vulgaris* aquaporins together with the tomato aquaporins identified by Reuscher et al. (2013). Prediction of transmembrane domains was performed using TMHMM server v.2.0, using default parameters (<http://www.cbs.dtu.dk/services/TMHMM/>) (Krogh et al. 2001). Prediction of sub-cellular localization was performed using both WoLF PSORT (Horton et al. 2007) and Plant-mPLoc (Chou and Shen 2010) algorithms with default parameters.

Expression analysis

For expression analysis in different organs, the FPKM (fragments per kilobase of transcript per million mapped reads) values for each aquaporin were extracted from the common bean genes.fpkms_tracking file, available in Phytozome. From this expression table, the FPKM values <1 were excluded and treated as a lack of expression of the corresponding gene (FPKM = 0). For analysis of aquaporin expression in different wild and domesticated *P. vulgaris* accessions, raw RNA-sequencing reads from Bellucci et al. (2014) (SRP028116) were downloaded from Short Read Archive (SRA) in NCBI using the R package (www.r-project.org) SRadb (Zhu et al. 2013). Reads were quality-filtered with Sickel (<https://github.com/najoshi/sickle>) using default parameters, and only reads without any uncertain called bases (Ns) were retained for subsequent alignment. Reads were aligned on the *P. vulgaris* genome sequence using TopHat (Trapnell et al. 2009) with a maximum of five mismatches, a maximum gap length of 3 and a maximum of one mismatch in the anchor region of a spliced alignment. Transcripts expression, represented as FPKM, were estimated using cufflinks (Trapnell et al. 2010) using upper-quartile normalization (-N) and

multi-read correction (-u) parameters. For subsequent analyses, only the FPKM value of each gene model, without considering splicing isoforms, was taken into account. For co-expression analysis, the Pearson correlation coefficient (r) based on FPKM was calculated.

To analyze gene expression of aquaporins in response to drought stress, all the *P. vulgaris* ESTs were downloaded from NCBI. We then divided all the ESTs in drought-related (i.e., libraries prepared from plants under drought stress) and background ESTs (all ESTs), and aligned them with BLASTN to *P. vulgaris* coding DNA sequences (CDSs). BLAST searches were performed with an e value cutoff of $1 \times e^{-10}$ and only the best hit for each EST was considered for subsequent analysis. Only aquaporins identified by both datasets were taken into account for differential expression analysis. Expression was estimated by counting the sequences aligning to a specific CDS. Differential expression was evaluated with a χ^2 test, based on a 2×2 contingency table of alignment counts. For each χ^2 test, *p* values were corrected with 10^3 Monte Carlo simulations of *p* values. Expression between the two datasets was normalized by dividing the number of ESTs aligning to a specific aquaporin by all the aligned ESTs in the dataset. Differential expression was expressed as the ratio between the two normalized values.

Integration of previously characterized aquaporins and BARCBean6K_3 SNP markers

Protein sequences of the common bean aquaporins previously characterized by Aroca et al. (2006), and Montalvo-Hernández et al. (2008) were downloaded from the NCBI nucleotide database (<http://www.ncbi.nlm.nih.gov/nucleotide>) using the respective GenBank ID (U97023, AY995196, AY995195, DQ087217, and DQ087218). The sequences were blasted against the *P. vulgaris* proteome using the BLASTP program, with default parameters, in the Phytozome platform (Goodstein et al. 2012). Since the genomic positions of the BARCBean6K_3 SNP loci are known (Bello et al. 2014), the BEDTools suite (Quinlan and Hall 2010) was used for calculating the distance between common bean aquaporins and BARCBean6K_3 SNP markers. To identify drought-related SNP markers in the BARCBean6K_3 chip, we chose the markers nearest to drought QTLs identified by Mukeshimana et al. (2014).

Results

Aquaporin identification and analysis

Analysis of the available *P. vulgaris* genome identified 42 putative aquaporin-encoding genes located on all but one

chromosome, in a non-uniform manner. Chromosome 3 (Chr 03) contained one aquaporin (Phvul.003G040100). In contrast, Chromosomes 1 and 11 contained six aquaporins each. Chromosome 6 had two members of the NIP sub-family exclusively. Manual checking of multiple protein alignment of the putative identified aquaporins showed that Phvul.003G040100 has a complete deletion of the first NPA domain (Supplementary File S1). Analysis of RNA-Sequencing read coverage using the jbrowse plugin on Phytozome showed a low log-coverage (<1) for this gene (Phytozome jbrowse RNA-Sequencing log-coverage link in Supplementary File S1). Moreover, analysis of expression with the Phytomine tool on Phytozome showed a FPKM of 0 in every organ analyzed (Phytomine FPKM expression link in Supplementary File S1). Thus, this protein was removed for further analysis because it is probably a mis-annotation or a pseudo-gene sequence. With the removal of this sequence, the total number of aquaporins in the *P. vulgaris* genome is 41, with no aquaporin located on Chr 03 (Fig. 1). The number of aquaporins identified in common

bean is higher than that identified in *Arabidopsis* and maize, but lower compared to species like *P. trichocarpa* and tomato. The list of the identified aquaporins, together with basic statistics on primary protein sequences, is reported in Table 1. Protein length ranged from 235 (Phvul.001G108800) to 324 residues (Phvul.008G041100), with a median of 272 amino acids per protein. Prediction of transmembrane domains showed six transmembrane domains (TMDs) for the majority of aquaporins (32 of 41, 78 %), except for six aquaporins having five predicted TMDs and three having seven predicted TMDs (Table 1). Multiple sequence alignment of the 6th helix (H6) region does not show any gap in this region for protein predicted to have 5 TMDs, except for PvPIP1;4, which shows a shorter C-term (Fig. 2). However, PvPIP1;4 is expressed in almost every organ and genotype analyzed, except in green mature pods and in the leaves of genotype PI325677 (Supplementary File S2). The two different sub-cellular localization prediction tools identified the same localization for 23 aquaporins (56 %), locating them mainly in the plasma

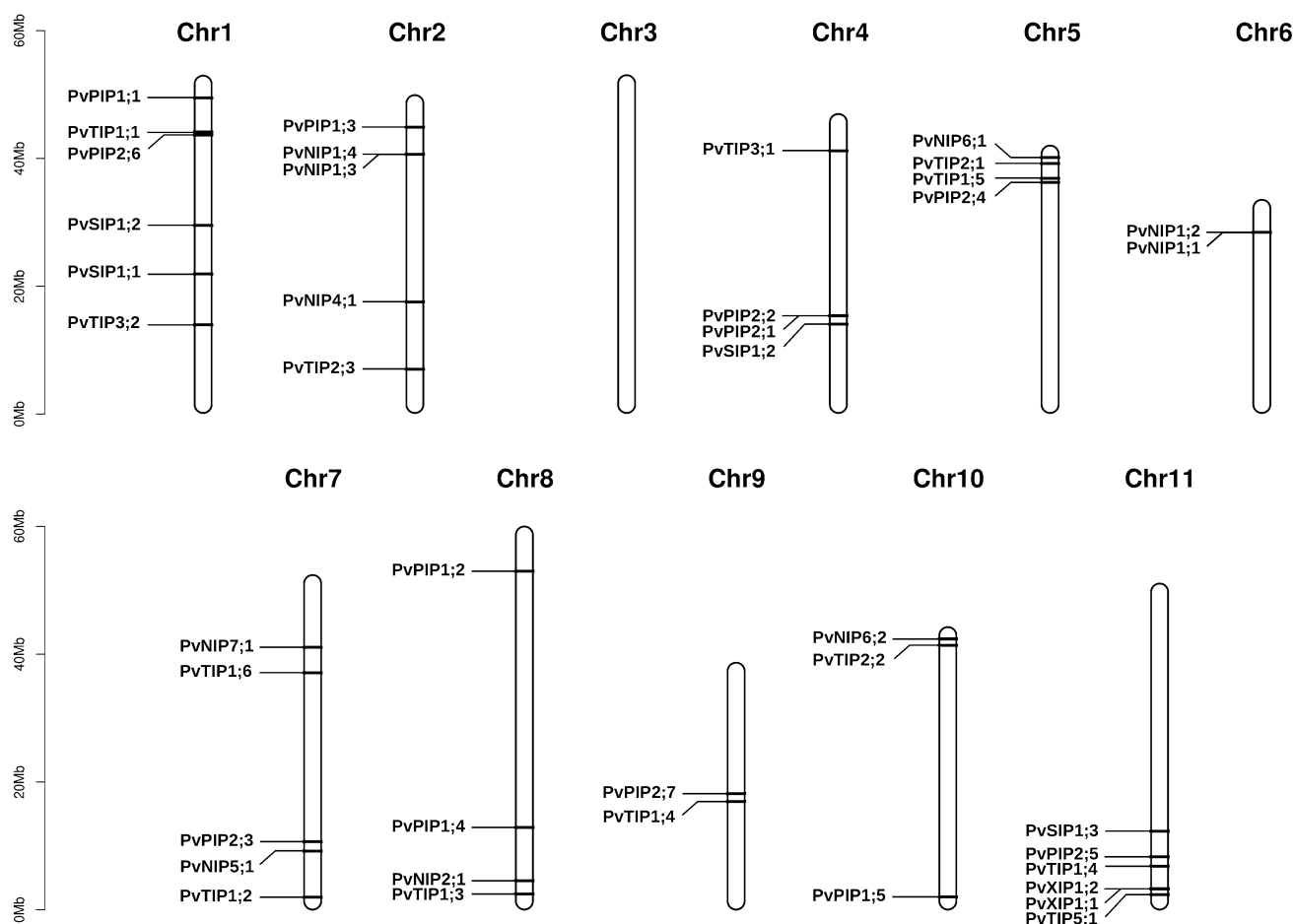


Fig. 1 Chromosomal localization of the 41 full-length coding aquaporin sequences in the common bean genome. Position determined in bp based on the *Phaseolus vulgaris* reference sequence

Table 1 Nomenclature and analysis of primary structure of common bean aquaporins

	Gene name	Locus	AA	TMD	WoLF PSORT ^a	Plant-mPLOC ^a
PIP	PvPIP1;1	Phvul.001G241200	289	6	Plas	Plas
	PvPIP1;2	Phvul.008G226000	289	5	Plas	Plas
	PvPIP1;3	Phvul.002G290400	287	6	Plas	Plas
	PvPIP1;4	Phvul.008G106400	249	5	Plas	Plas
	PvPIP1;5	Phvul.010G004500	287	6	Plas	Plas
	PvPIP2;1	Phvul.004G082600	285	6	Plas	Plas
	PvPIP2;2	Phvul.004G082700	285	6	Plas	Plas
	PvPIP2;3	Phvul.007G094600	287	6	Plas	Plas
	PvPIP2;4	Phvul.005G135300	287	6	Plas	Plas
	PvPIP2;5	Phvul.011G079300	285	6	Plas	Plas
	PvPIP2;6	Phvul.001G177000	286	6	Plas	Plas
	PvPIP2;7	Phvul.009G118900	278	6	Plas	Plas
TIP	PvTIP1;1	Phvul.001G181100	250	7	Cyto	Vacu
	PvTIP1;2	Phvul.007G010800	252	6	Cyto/plas	Vacu
	PvTIP1;3	Phvul.008G014900	252	6	Cyto	Vacu
	PvTIP1;4	Phvul.011G067200	252	6	Vacu	Vacu
	PvTIP1;5	Phvul.005G141100	238	6	Vacu	Vacu
	PvTIP1;6	Phvul.007G151600	253	6	Mito	Vacu
	PvTIP2;1	Phvul.005G170300	248	6	Plas	Vacu
	PvTIP2;2	Phvul.010G147100	248	7	Vacu	Vacu
	PvTIP2;3	Phvul.002G057500	249	6	Plas	Vacu
	PvTIP3;1	Phvul.004G136000	256	6	Mito	Vacu
	PvTIP3;2	Phvul.001G080300	254	6	Vacu	Vacu
	PvTIP4;1	Phvul.009G108800	246	6	Vacu	Vacu
	PvTIP5;1	Phvul.011G015200	252	7	Chlo	Plas/vacu
	PvNIP1;1	Phvul.006G170900	260	6	Plas	Plas
	PvNIP1;2	Phvul.006G171000	273	6	Plas	Plas
NIP	PvNIP1;3	Phvul.002G242200	272	5	Vacu	Plas
	PvNIP1;4	Phvul.002G242300	272	6	Vacu	Plas
	PvNIP2;1	Phvul.008G041100	324	6	Mito	Plas
	PvNIP4;1	Phvul.002G093700	261	6	Vacu	Plas/vacu
	PvNIP5;1	Phvul.007G084000	299	5	Plas	Plas
	PvNIP6;1	Phvul.005G182400	303	6	Plas	Plas
	PvNIP6;2	Phvul.010G160200	301	6	Plas	Plas
	PvNIP7;1	Phvul.007G178300	297	6	Plas	Plas
	PvSIP1;1	Phvul.001G097000	248	6	Vacu	Plas
SIP	PvSIP1;2	Phvul.004G077700	247	6	Vacu	Plas/vacu
	PvSIP1;3	Phvul.011G102700	239	5	Plas	Plas/vacu
	PvSIP2;1	Phvul.001G108800	235	5	Vacu	Plas
	PvXIP1;1	Phvul.011G025700	308	6	Cyto	Plas
XIP	PvXIP1;2	Phvul.011G025800	280	6	Chlo	Plas

Chlo chloroplast, *Cyto* cytosol, *Mito* mitochondria, *Plas* plasma membrane, *Vacu* vacuolar membrane

^a Best possible sub-cellular localization of common bean aquaporins based on two different algorithms

membrane and tonoplast. For the other 18 aquaporins, Plant-mPLOC predicted a localization mainly in the plasma membrane or tonoplast; in contrast, WoLF PSORT predicted several different sub-cellular localizations, including chloroplast, cytoplasm, and mitochondria (Table 1).

Phylogenetic and tandem duplication analysis of aquaporin family

A phylogenetic analysis combining common bean aquaporins with those of *Arabidopsis* and *P. trichocarpa* clearly

Fig. 2 Multiple protein alignment of the *P. vulgaris* aquaporins in the region surrounding the 6th helix (H6). The gray rectangle and the black arrow highlight the position and orientation of H6. The number in the upper left side indicates the starting position of the multiple alignment

						H6	
	301						
PvPIP2;3	-VTGTGINPA	RSLGAAVIYN	KDKPWDDHWI	F'W'G'P'F'IGAA	I'AA'F'Y	-----	
PvPIP2;7	-ITGTGINPA	RSFGAAVIYN	DAK'V'W'DDHWI	F'W'G'P'F'VGAL	A'AA'Y	-----	
PvPIP2;5	-VTGTGINPA	RSFGAAVIYN	KEKI'W'DDQWI	F'W'G'P'I'IGAA	V'AA'F'Y	-----	
PvPIP2;1	-VTGTGINPA	RSLGAAVIYN	KEKA'W'DDQWI	F'W'G'P'F'IGAA	I'AA'F'Y	-----	
PvPIP2;2	-VTGTGINPA	RSLGAAVIYN	KEKA'W'DDQWI	F'W'G'P'F'IGAA	I'AA'F'Y	-----	
PvPIP2;4	-ITGTGINPA	RSFGPAVIFN	NDKA'W'DDQWI	Y'W'G'P'F'IGAA	V'AA'F'Y	-----	
PvPIP2;6	-VTGTGINPA	RSLGPAVIFN	NQKA'W'DDQWM	F'W'G'P'F'VGAG	I'AA'V'Y	-----	
PvPIP1;3	-ITGTGINPA	RSLGAAI'Y'N	RDHA'W'DDQWI	F'W'G'P'F'IGAA	I'AA'LY	-----	
PvPIP1;5	-VTGTGINPA	RSLGAALIYN	KDYI'W'DEHWI	F'W'G'P'F'IGAA	V'AA'LY	-----	
PvPIP1;2	-ITGTGINPA	RSLGAAI'F'N	KDLG'W'DDHWI	F'W'G'P'F'IGAA	I'AA'LY	-----	
PvPIP1;1	-ITGTGINPA	RSLGAAI'F'N	KDLG'W'DDHWI	F'W'G'P'F'VGAA	I'AA'LY	-----	
PvNIP1;2	-ITGASMPA	RSLGPAIV--	-HNE'Y'RG'WI	Y'LV'SP'TLGAV	A'GA'W'A	-----	
PvNIP1;4	-ITGASMPA	RSLGPAIV--	-HNE'F'RG'WI	Y'LV'SP'TFGAF	V'GA'W'V	-----	
PvNIP6;1	-VSGGSMNPV	RTLGP'AI'A	-ANN'Y'KAI'W	Y'LV'AP'V'LGAL	A'G'AG'T	-----	
PvNIP1;3	-VTGASMPV	RSLGPAFV--	-HSE'Y'TGI'W	Y'IL'AP'ILGAV	V'GA'W'V	-----	
PvNIP5;1	-TSGGSMNPV	RTLGP'AVA	-AGN'Y'KHI'W	Y'LV'AP'T'LGAL	A'G'AG'V	-----	
PvNIP6;2	-ATGSSMPV	RTLGP'AI'A	-ANN'Y'KGI'W	Y'LI'AP'IFGTL	C'G'AG'A	-----	
PvNIP1;1	-ITGASMPA	RSLGP'AIL--	-HHE'Y'RG'WI	Y'M'V'SP'IVGAL	A'AI'W'L	-----	
PvPIP1;4	-ITGTGINPA	RSLVSLII--	-----I	L*	-----	-----	
PvTIP2;2	-FSGGSMNPA	RSFGPAV--	-SGD'F'HDNWI	Y'W'G'P'LIGGG	I'AG'LI	-----	
PvTIP2;1	-FSGGSMNPA	RSFGPAIV--	-NGD'F'HNNWI	Y'W'G'P'LVGGG	I'AG'LV	-----	
PvNIP4;1	-VSGASMPA	RSLGP'ALV--	-MHV'Y'SGF'W	Y'IV'G'P'F'VGAL	I'G'AT'C	-----	
PvTIP2;3	-FSGGSMNPA	RSFGPAV--	-SGN'F'ADNWI	Y'W'G'P'LIGGG	I'AG'LI	-----	
PvTIP3;2	-FDGACMPA	RAFGP'ALV--	-GWR'W'QYHWI	F'W'G'P'LLGAA	I'AA'V'L	-----	
PvTIP3;1	-FDGACMPA	LAFGP'SLV--	-GWQ'W'QHWI	F'W'G'P'LLGAA	I'AA'LV	-----	
PvTIP1;1	-FSGAAMPNPA	VTFGP'AVV--	-SWI'W'INNWI	Y'W'G'P'LIGGG	I'AG'LI	-----	
PvTIP1;4	-FDGASMPA	VSFGP'AVV--	-SGI'W'ANHWV	Y'W'G'P'LIGSA	I'AA'V'V	-----	
PvTIP1;6	-FDGGSMPA	AAFGP'ALV--	-GWN'W'ENHWV	Y'W'G'P'LVGGG	I'AG'F'L	-----	
PvTIP1;2	-FDGASMPA	VSFGP'ALV--	-SWS'W'THHWV	Y'W'G'P'FIGSA	A'AA'LL	-----	
PvTIP1;3	-FDGASMPA	VSFGP'AVV--	-TWQ'W'INHWV	Y'W'G'P'FIGAA	I'AA'I'I	-----	
PvTIP4;1	-YSAASMPA	RSFGPAIV--	-TGN'W'IDHWV	Y'W'G'P'LIGGG	I'AG'F'I	-----	
PvTIP1;5	-FDGASMPA	VSFGP'AVV--	-SWI'W'ANHWV	Y'W'G'P'FAGAA	I'AA'V'V	-----	
PvNIP2;1	-ISGGSMNPA	RTLGP'AI'A	-TSS'Y'KGLWL	Y'F'V'G'P'ITGAI	I'AA'W'S	-----	
PvNIP7;1	-VSGGSMNPA	RSLGP'AIL--	-SWK'F'KNIWI	Y'M'V'AP'SGGAV	A'G'AA'M	-----	
PvTIP5;1	-FSGGSMNPA	CAFGS'ATI--	-AGS'F'RNQAV	Y'W'G'P'LIGAT	I'AG'LL	-----	
PvSIP1;1	-YTGPAMPNPA	NAFGW'AYLNN	WHNT'W'DQLYV	Y'W'IC'P'FAGAI	I'AA'W'L	-----	
PvSIP1;2	-YTGPAMPNPA	FAFGW'AYF'GK	WHDK'W'DHFYV	Y'W'IC'P'FVGS	I'AA'W'L	-----	
PvSIP1;3	-FTGPSPNPA	NAFGW'AFINN	KHNT'W'EQFYV	Y'W'IC'P'FIGAS	S'AA'F'I	-----	
PvSIP2;1	-LTGGCMNPA	SVMGW'AYARG	DHIT'K'EHFFV	Y'W'LA'P'IEATI	I'AW'W'T	-----	
PvXIP1;1	GYAGV'GLNPA	RCLGP'ALLHG	-GSL'W'NGHWI	F'W'G'P'FLACI	I'YYS'V'SIILP		
PvXIP1;2	GYAGV'GLNPA	RCLGP'ALLHG	-GSL'W'HGHV	F'W'G'P'FLACM	L'Y'Y'V'V'SINLP		

grouped this protein family into five sub-families consisting of 12 PIPs, 13 TIPs, 10 NIPs, 4 SIPs, and 2 XIPs (Fig. 3; Table 1). The TIPs sub-family was the biggest one with 13 members divided into five groups (TIP1 to TIP5), six members in the TIP1 group, three in TIP2, two in TIP3 and one each in groups TIP4 and TIP5. The PIP sub-family was divided into PIP1, with 5 members, and PIP2, with 7 members. NIPs were divided into NIP1, with 3 members, NIP6, with 2 members, and NIP2, 4, 5 and 7, with one member each. SIPs were divided into SIP1, with 3 members, and SIP2, with one member. Only one group of XIPs was identified, with 2 members.

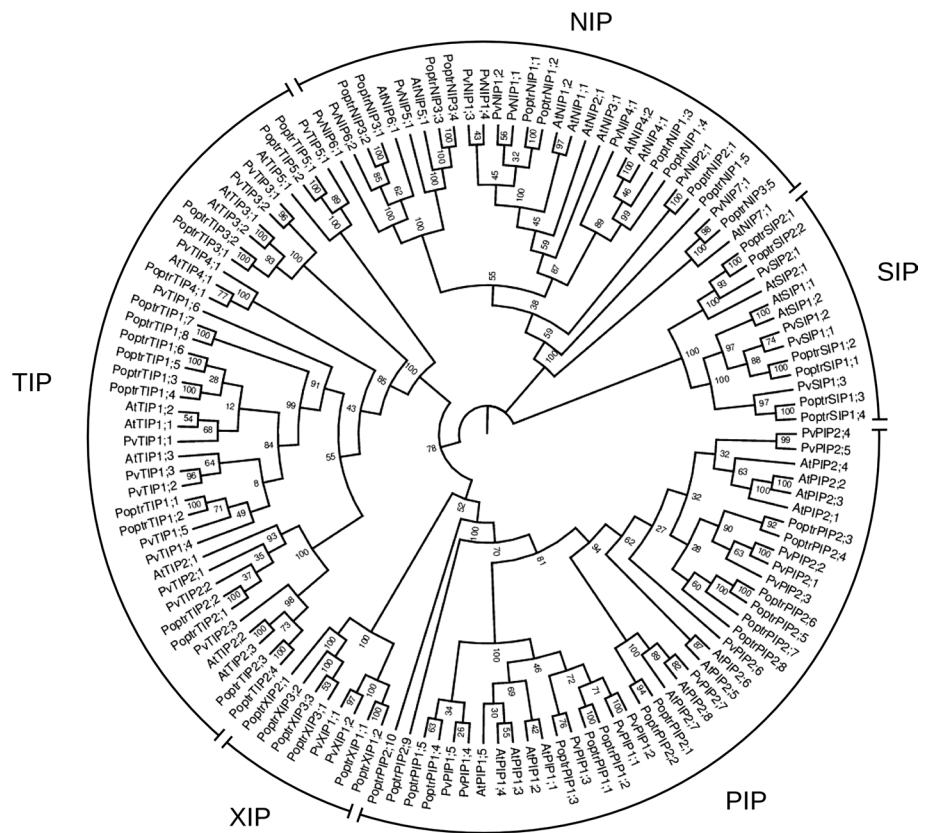
A pair of putatively tandem duplicated genes was detected for two PIP2 aquaporins, four NIP1 (two pair of duplicated genes), and the two members of the XIP sub-family. A pairwise protein sequence analysis showed family-specific percentage of identity among the putatively duplicated pairs. In particular, PvPIP2;1 and PvPIP2;2

showed ~98 % identity, the two pairs of NIPs (PvNIP1;1 and PvNIP1;2, PvNIP1;3 and PvNIP1;4) ~71 % and the two XIPs less than 60 % (Supplementary File S3).

Analysis of exon–intron structure

The complete transcript annotation of *P. vulgaris* genome enables the analysis and comparison of the structural features of aquaporins in different families. Within each aquaporin sub-family, the number, order and length of exons were almost conserved; in contrast, the introns and UTRs showed variable lengths and patterns (Fig. 4). The majority of the members of the PIP sub-family were characterized by four exons of similar length, except PvPIP1;4, which showed a short 4th exon without a 3'UTR, and PvPIP2;6, which had only three exons. Most of the members of the TIP sub-family showed three exons, except PvTIP1;1 and PvTIP1;6, with two, and PvTIP1;5, with four exons. In

Fig. 3 Bio Neighbor-Joining (BioNJ) phylogenetic tree (1000 bootstraps), based on protein alignment of aquaporins from *Arabidopsis*, *P. trichocarpa* and *P. vulgaris*. Numbers on branches represent bootstrap values. The abbreviation of the species is as follows: *At Arabidopsis*, *Poptr P. trichocarpa*, *Pv P. vulgaris*



addition, PvTIP1;2, PvTIP1;5 and PvTIP5;1 did not have UTRs at both 5' and 3' ends. The genes in the NIP sub-family showed mainly five exons, except for PvNIP2;1, PvNIP5;1 and PvNIP6;1, which exhibited four. In this family, PvNIP4;1 did not have any UTRs and PvNIP7;1 showed a longer 5' UTR compared to other aquaporins in the same family. SIP genes showed three exons, except PvSIP1;3 that had only one exon and no introns. The two members of the XIP sub-family showed completely different features, with PvXIP1;1 having two exons, and PvXIP1;2 having one exon without any UTRs.

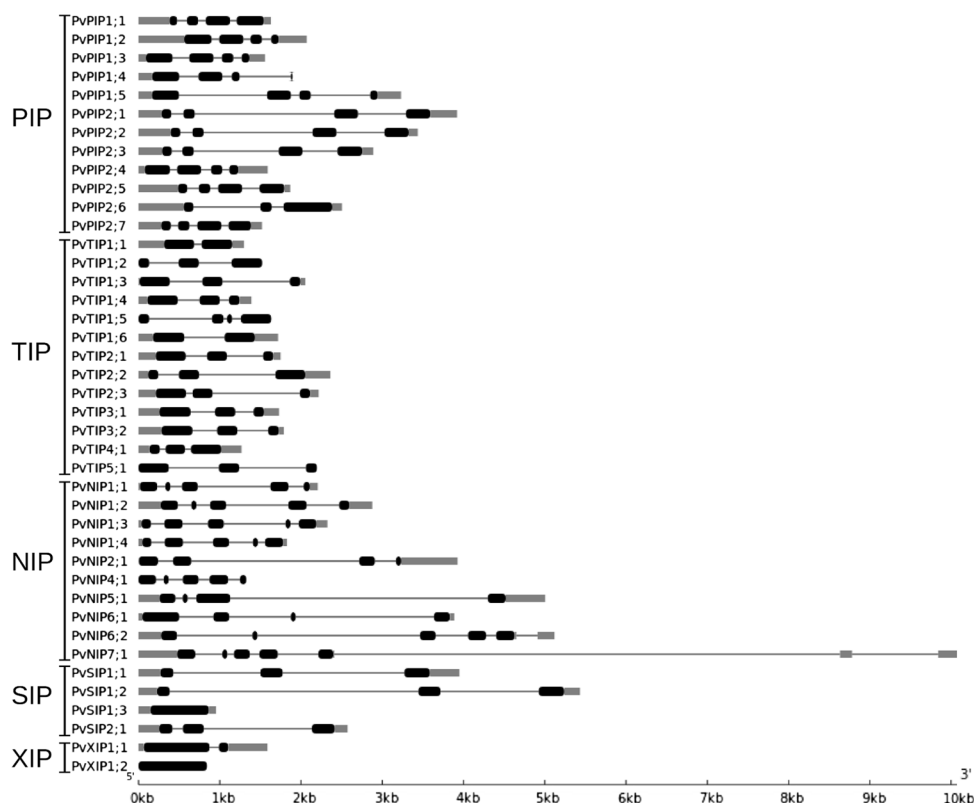
Analysis of conserved and substrate-specific residues in aquaporin proteins

To understand the possible physiological role and substrate specificity of common bean aquaporins, we identified and analyzed the NPA motifs and ar/R selectivity filter sequences (Table 2). Both in the PIP and TIP sub-families, the two NPA domains are conserved and showed the typical Asp-Pro-Ala residues of aquaporins. In the NIP sub-family, the first NPA showed the same sequence as in PIPs and TIPs, except for PvNIP5;1, where the alanine is replaced by a serine residue. The second NPA motif showed an alanine to valine substitution in four NIPs (PvNIP1;3, PvNIP1;5,

PvNIP6;1 and PvNIP6;2). SIP aquaporins showed a second NPA motif completely conserved with the other sub-families, instead all of the first NPA motifs show the replacement of alanine by threonine (PvSIP1;1 and PvSIP1;2), serine (PvSIP1;3) or leucine (PvSIP2;1). The XIP sub-family showed a second, completely conserved NPA, while the alanine is changed to isoleucine (PvXIP1;1) or leucine (PvXIP1;2) in the first NPA.

The ar/R positions showed an increased family-specific sequence compared to the two NPA motifs. In all PIPs, this selectivity filter showed the conserved residues, observed in other species, phenylalanine in H2, histidine in H5, threonine in LE1, and arginine in LE2. In the TIP sub-family, the ar/R is formed by histidine/asparagine/serine in H2; isoleucine/valine in H5, alanine/glycine in LE1 and valine/alanine/arginine/leucine/cysteine in LE2. In the NIP sub-family, this selection filter is constituted by tryptophan/alanine/threonine/asparagine in H2, valine/serine/isoleucine in H5, alanine/glycine/serine in LE1 and arginine in LE5. The SIP sub-family showed phenylalanine/alanine/tyrosine in H2, threonine/lysine in H5, proline/glycine in LE1, and asparagine/phenylalanine/serine in LE2. The XIPs sub-family is quite homogeneous in these sites showing valine in H2 and LE1, methionine in H5, and asparagine in LE2.

Fig. 4 Gene structure of the 41 full-length coding aquaporin sequences in common bean. UTRs are represented by gray rectangles, CDSs by black round-corner rectangles, and introns by black lines



Expression analysis

Gene expression data in different organs with RNA-sequencing are available from Phytozome (see Materials and Methods) and were downloaded for characterizing the expression in different organs of the identified aquaporins. A heat map showing aquaporin expression is shown in Fig. 5a. Thirty-six aquaporins (~88 %) were expressed in at least one organ analyzed, and 17 of them (~41 %) were expressed in all organs, including 10 PIPs, 4 TIPs, 2 SIPs, and 1 NIP. The number of aquaporins expressed in different organs range from 25 to 33, with the highest number in young pods (33, ~80 %) and flower buds (30, ~73 %), and the lowest in green mature pods and leaves (25 each, ~61 %). Among all the *P. vulgaris* aquaporins, PvTIP1;1 showed the highest expression level in all the organs analyzed; in contrast, five aquaporins (PvTIP1;2, PvTIP3;2, PvTIP5;1, PvNIP1;1 and PvXIP1;2) were not expressed in any of these organs. The aquaporins with the highest mean expression (>90 % percentile) across all organs were PvTIP1;1 (mean FPKM = 2303.06), PvPIP1;2 (mean FPKM = 672.22), PvPIP2;4 (mean FPKM = 589.01) and PvPIP2;7 (mean FPKM = 448.68).

Analysis of expression in different genotypes showed a similar expression pattern observed within different tissues (Supplementary file S4), with PvTIP1;1 having the highest expression of all aquaporins in almost all genotypes.

The aquaporins with the highest mean expression (>90 % percentile) across all genotypes were PvTIP1;1 (mean FPKM = 2291.62), PvPIP2;4 (mean FPKM = 1363.05), PvPIP1;2 (mean FPKM = 1244.26) and PvPIP1;1 (mean FPKM = 781.36).

Since hetero-tetramerization between aquaporin isoforms, especially between members of PIP1 and PIP2 sub-families, have been previously observed, we performed a co-expression analysis among the different genotypes for identifying possible interacting proteins. This analysis showed that the majority of common bean aquaporins did not have any correlation at the expression level, except for a few members of PIPs sub-family (Fig. 5b). Among these co-expressed proteins, we found three pairs (PvPIP1;4 and PvPIP2;7, PvPIP1;4 and PvPIP2;3, PvPIP1;3 and PvPIP2;7) with a Pearson correlation coefficient >0.9 (Supplementary file S5).

Expression analysis in response to drought stress was performed using EST databases available in NCBI. ESTs for 28 aquaporins (~68 %) were identified in at least one of the datasets, while in the drought-related dataset 254 ESTs identify only 12 aquaporins (~30 %). The aquaporins with the highest expression (based on the number of ESTs aligning to them) were PvTIP1;1, PvPIP1;2, PvPIP2;4 and PvPIP2;5. Differential expression analysis based on χ^2 test showed a significant differential expression ($p < 0.01$) for PvTIP1;1, with a relative increase of ~1.6 fold in comparison to the control dataset (Supplementary file S6).

Table 2 NPA motifs and ar/R selectivity filter of common bean aquaporins

Sub-family	Name	NPA motifs		ar/R Selectivity filter			
		I	II	H2	H5	LE1	LE2
PIP	PvPIP1;1	NPA	NPA	F	H	T	R
	PvPIP1;2	NPA	NPA	F	H	T	R
	PvPIP1;3	NPA	NPA	F	H	T	R
	PvPIP1;4	NPA	NPA	F	H	T	R
	PvPIP1;5	NPA	NPA	F	H	T	R
	PvPIP2;1	NPA	NPA	F	H	T	R
	PvPIP2;2	NPA	NPA	F	H	T	R
	PvPIP2;3	NPA	NPA	F	H	T	R
	PvPIP2;4	NPA	NPA	F	H	T	R
	PvPIP2;5	NPA	NPA	F	H	T	R
	PvPIP2;6	NPA	NPA	F	H	T	R
	PvPIP2;7	NPA	NPA	F	H	T	R
TIP	PvTIP1;1	NPA	NPA	H	I	A	V
	PvTIP1;2	NPA	NPA	H	I	A	V
	PvTIP1;3	NPA	NPA	H	I	A	V
	PvTIP1;4	NPA	NPA	H	I	A	V
	PvTIP1;5	NPA	NPA	H	I	A	V
	PvTIP1;6	NPA	NPA	N	I	G	A
	PvTIP2;1	NPA	NPA	H	I	G	R
	PvTIP2;2	NPA	NPA	H	I	G	R
	PvTIP2;3	NPA	NPA	H	I	G	R
	PvTIP3;1	NPA	NPA	H	I	A	L
	PvTIP3;2	NPA	NPA	H	I	A	R
	PvTIP4;1	NPA	NPA	H	I	A	R
	PvTIP5;1	NPA	NPA	S	V	G	C
NIP	PvNIP1;1	NPA	NPA	W	V	A	R
	PvNIP1;2	NPA	NPA	W	V	A	R
	PvNIP1;3	NPA	NPV	W	V	A	R
	PvNIP1;4	NPA	NPA	W	V	A	R
	PvNIP2;1	NPA	NPA	A	S	G	R
	PvNIP4;1	NPA	NPA	W	V	A	R
	PvNIP5;1	NPS	NPV	A	I	G	R
	PvNIP6;1	NPA	NPV	T	I	G	R
	PvNIP6;2	NPA	NPV	N	I	S	R
	PvNIP7;1	NPA	NPA	A	V	G	R
SIP	PvSIP1;1	NPT	NPA	F	T	P	N
	PvSIP1;2	NPT	NPA	F	T	P	F
	PvSIP1;3	NPS	NPA	A	T	P	N
	PvSIP2;1	NPL	NPA	Y	K	G	S
XIP	PvXIP1;1	NPI	NPA	V	M	V	R
	PvXIP1;2	SPV	NPA	V	M	V	R

Integration of previously characterized aquaporins and BARCBean6K_3 SNPs markers

Protein sequences of the aquaporins previously characterized by Aroca et al. (2006) and Montalvo-Hernández et al. (2008) were assigned to the current

gene nomenclature based on the best blast alignment. PvPIP1;2 is the best blast hit for U97023 (97.9 % identity), PvPIP1;3 for AY995196 (99.7 % identity), PvPIP2;5 for AY995195 (98.6 % identity), and PvTIP2;3 for both DQ087217 (100 % identity) and DQ087218 (100 % identity).

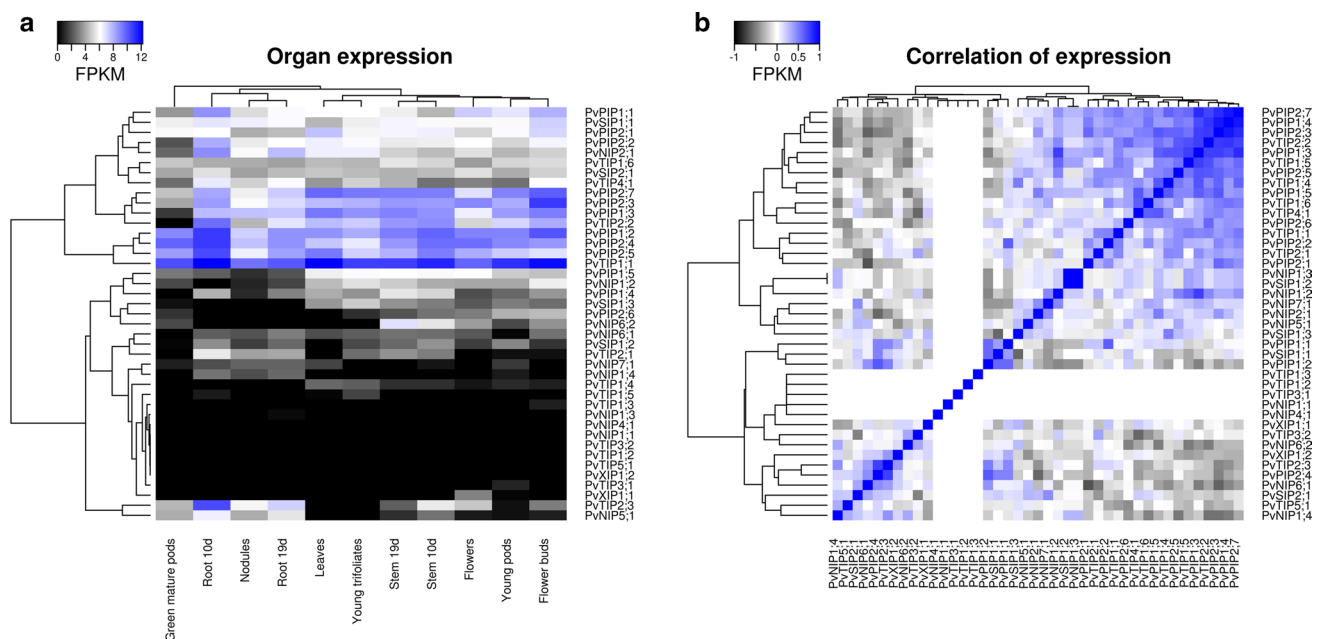


Fig. 5 Expression and co-expression profiles of the 41 aquaporins in common bean. **a** Expression heatmap in different organs. The expression values of each aquaporin are $\log_2$ transformed. 10d and 19d represent the expression in the corresponding organs at 10 or 19 days

Nine (22 %) of the common bean aquaporins are the nearest annotated locus to eight different SNP markers, with PvPIP2;1 being the closest locus to both ss715641384 and ss715650422 (Supplementary file S7). In particular, three SNP markers are located inside gene sequences of three different aquaporins: ss715641100 is located in an intron sequence of PvSIP1;2; in contrast, ss715645362 and ss715639327 are located in exon sequences of PvTIP2;1 and PvSIP1;3, respectively. The other aquaporins are located between 1500 and 13,000 bp from the closest SNP marker (Supplementary file S8). Recently, Mukeshimana et al. (2014) identified several Quantitative Trait Loci (QTLs) associated with drought stress in common bean using the BARCBean6K_3 genotyping platform. None of the common bean aquaporins was the closest annotated locus to identified drought QTLs. The distance between the QTLs and the closest aquaporins ranged from 0.4 to 14 Mb (Supplemental file S6).

Discussion

Aquaporins are an abundant and highly divergent protein family in plants. Recent efforts in plant genome sequencing projects have enabled the identification and characterization of the complete set of aquaporins in several plant species (Chaumont et al. 2001; Johanson et al. 2001; Gupta and Sankararamakrishnan 2009; Reuscher et al. 2013; Tao

et al. 2014). These analyses could be extremely helpful in the identification of physiologically important aquaporins for a specific species. In addition, in silico analyses of aquaporin protein family could guide other researchers in hypothesis generation and testing, facilitating in depth functional characterization of this protein family.

The recent release of *P. vulgaris* genome (Schmutz et al. 2014) allowed the identification and characterization of the entire aquaporin family in this study. For the best of our knowledge, this is the first genome-wide analysis of aquaporin family in common bean, and one of the first genome-wide characterization of complete gene family in this species. We identified 41 full-length aquaporin-coding sequences in the common bean genome, with 12 genes belonging to PIPs sub-family, 13 to TIPs, 10 to NIPs, four to SIPs, and two to XIPs. The number of aquaporins identified in the common bean is higher than those identified in species like *Arabidopsis* and maize, but lower compared to species like *P. trichocarpa* and tomato. This high number of aquaporins could be interpreted as a possible functional redundancy and/or sub- or neo-functionalization of this protein family in common bean. A note of caution should be introduced here. The distribution and count of aquaporin sequences was based on the current reference sequence of common bean (Schmutz et al. 2014), which originated in the Andean landrace G19833. Based on the divergence between the Andean and Mesoamerican gene pools, which includes partial reproductive isolation (Koinange and Gepts

1992), a similar analysis should be conducted with Mes-america reference sequences, once these become available (http://phasibeam.crg.eu/wiki/Main_Page; <http://www.beangenomics.ca/>).

Analysis of chromosomal distribution of aquaporins showed the existence of tandem duplications within aquaporin families. Gene duplication is an important mechanism for increasing genetic variability and creating novel genes (Magadum et al. 2013). In plants, this mechanism is a major force in the increase of gene family size and in functional expansion of genomes (Severing et al. 2009; Magadum et al. 2013). Duplicated genes could undergo pseudogenization, gene conservation and sub- or neo-functionalization (Lynch and Conery 2000; Flagel and Wendel 2009).

The high similarity at both protein and gene structure levels between PvPIP2;1 and PvPIP2;2 suggests a recent tandem duplication and, possibly, a redundant function of these two genes in common bean. This redundant function could also be inferred from similar expression patterns in different organs. In the NIP sub-family, two pairs of putatively tandem duplicated genes were identified: PvNIP1;1–PvNIP1;2 and PvNIP1;3–PvNIP1;4. These pairs of genes showed a similar gene structure, but displayed just ~70 % similarity at the protein level and divergent expression patterns. In particular, PvNIP1;1 could have undergone pseudogenization since it was not expressed in any organ. PvNIP1;3 is only slightly expressed in 19-day-old roots, while PvNIP1;4 is highly expressed in both 10- and 19-day-old roots. The most striking difference among this pair of putatively duplicated genes is in the two members of the XIP sub-family. Even though they are adjacent on Chromosome 11 they showed a low level of protein sequence similarity, different expression patterns, and a completely different gene structure. PvXIP1;1 is expressed only in flowers, while PvXIP1;2 is not expressed in any organ.

An essential feature of aquaporins is the presence of two NPA motifs that form the water pore, essential for selective transport of water (Murata et al. 2000). In addition, the ar/R selectivity filter is essential in determining transport specificity of aquaporins. Point mutations or other sequence variations in these residues confer different substrate specificities to aquaporins (Beitz et al. 2006; Hub and de Groot 2008; Mitani-Ueno et al. 2011).

All *P. vulgaris* PIPs aquaporins showed the characteristic double NPA motif in their protein sequences. In addition, the ar/R selectivity filter is highly conserved and typical of water-transporting aquaporins (F/H/T/R). The same ar/R selectivity filter sequence is shared by the PIP sub-family in different plant species such as *Z. mays*, *A. thaliana*, *P. trichocarpa*, *S. lycopersicum*, *G. max* and *B. rapa* (Chau-mont et al. 2001; Johanson et al. 2001; Gupta and Sankararamakrishnan 2009; Reuscher et al. 2013; Zhang et al.

2013; Tao et al. 2014). Several researchers hypothesize a central role in root water absorption for plant PIP aquaporins, suggesting that this protein could also actively regulate root and leaf hydraulics (Maurel et al. 2008; Li et al. 2014). In addition to water transport, different studies have shown that PIPs aquaporins from *N. tabacum*, *A. thaliana* and barley could facilitate the diffusion of CO₂ in mesophyll and could directly affect photosynthesis (Heckwolf et al. 2011; Flexas et al. 2006; Mori et al. 2014). The conservation of important transport residues (i.e., NPA motifs and ar/R residues) in common bean PIPs could suggest a similar role of these proteins in regulating water absorption, plant hydraulics and/or CO₂ diffusion. Expression analysis showed that among PIPs, PvPIP1;2 and PvPIP2;4 have the highest expression across all organs and in leaves of different genotypes. Thus, we suggest that these two aquaporins could have a central physiological role in regulating water movement and homeostasis in common bean.

Members of the PIP1 and PIP2 sub-families interact in vivo. This interaction determines an additive increase in water transport and the re-localization of PIP1 aquaporins in plasma membrane (Fetter et al. 2004; Zelazny et al. 2007, 2009). Co-expression network analysis is widely used in plant and other organisms for identifying and annotating novel gene functions (Usadel et al. 2009), but it showed to be less useful for identifying interacting proteins (Xulvi-Brunet and Li 2010; Giorgi et al. 2013). Despite this, our co-expression analysis showed that pairs of PIP1 and PIP2 members have a high Pearson correlation coefficient (>0.9), pointing to a possible interaction between these proteins pair.

All *P. vulgaris* TIP aquaporins showed the canonical double NPA motif and a group specific ar/R selectivity filter that were conserved across different species. TIP aquaporins are located mainly in vacuolar membranes and regulate osmotic potential and water flow across this essential plant sub-cellular compartment (Maurel et al. 2008; Li et al. 2014). The expression levels of TIPs were also associated with cell growth, cell expansion, and response to gibberellic acid (Ludevid et al. 1992; Phillips and Huttly 1994). Several experiments have shown the ability of TIPs to also transport a wide variety of small solutes, such as NH₄⁺, H₂O₂, and urea, besides water (Liu et al. 2003; Holm et al. 2005; Loque et al. 2005; Bienert et al. 2007). The conservation of ar/R residues and NPA motifs between TIPs of common bean and other species could suggest a conserved function for these proteins. Expression analysis showed the highest mean expression in all organs for PvTIP1;1, suggesting its possible central role in regulating water homeostasis and cellular osmosis in common bean. Since the homologs of PvTIP1;1 in *Arabidopsis* and *Panax ginseng* were related to plant growth (Phillips and Huttly 1994), or increased growth rate and biomass production

when expressed in transgenic plants (Peng et al. 2007), we suggest that PvTIP1;1 could be involved in regulation of plant growth in common bean as well.

The NIPs aquaporin sub-family was named by its first described member, the *G. max* nodulin 26 protein, identified in the symbiosome membranes of the nitrogen-assimilating root nodules (Wallace et al. 2006). This aquaporin sub-family has a wide range of substrate specificity such as water, glycerol, urea, boric acid, silicon, and lactic acid (Dean et al. 1999; Wallace and Roberts 2005; Ma et al. 2006; Takano et al. 2006; Choi and Roberts 2007). This specificity is based mainly on sequence differences in the ar/R selectivity filter that are used also for dividing this sub-family in three distinct subgroups (Wallace and Roberts 2005). The ar/R selectivity filter of *P. vulgaris* NIP1 group is typical of the subgroup I of plant NIP aquaporins. These proteins are characterized by the residues W/V/A/R in the ar/R filter, low water permeability, and the capability of transporting uncharged solutes like glycerol and formamide (Wallace and Roberts 2005). Due to these structural similarities, we suggest that common bean NIP1 group could also have similar transport specificity as previously demonstrated for homologous aquaporins from other species.

OsNIP2;1 is a rice silicon transporter characterized by a double NPA motif and a G/S/G/R ar/R selectivity filter (Ma et al. 2006). This aquaporin is also able to transport arsenite and boric acid when expressed in *Xenopus* oocytes. Mutation in the H5 position of the ar/R selectivity filter resulted in a loss of transport activity for these molecules (Mitani-Ueno et al. 2011). On the other hand, a G to A mutation in the H2 position of the ar/R selectivity filter does not affect the transport capability of this protein (Mitani-Ueno et al. 2011). Since PvNIP2;1 showed sequence similarities with OsNIP2;1, with a G to A mutation in the H2 position of the ar/R selectivity filter, we suggest that this protein could be involved in both silicon or boric acid homeostasis in common bean. AtNIP5;1 is an *Arabidopsis* boric acid transporter characterized by a NPS/NPV aqueous pore and an A/I/G/R selectivity filter (Takano et al. 2006). Mutations in the ar/R selectivity filter regions alter the transport capability of this protein. Since AtNIP5;1 and its common bean homolog, PvNIP5;1, share the same NPS/NPV aqueous pore and ar/R selectivity filter, we suggest that PvNIP5;1 could be involved in boron transport in common bean.

Several experiments showed differential expression of plant aquaporins in response to hormones and to environmental stresses, such as drought, cold, and salinity (Li et al. 2014). This differential expression could be plant-, aquaporin-, or stress-specific. Differential expression of aquaporins in response to different environmental stresses has been comprehensively evaluated in several plant species such as *Arabidopsis*, maize, poplar and *B. rapa* in the past years (Alexandersson et al. 2005; Zhu et al. 2005; Cohen

et al. 2013; Tao et al. 2014). Regarding common bean, few aquaporins have been characterized to date. Aroca et al. (2006) cloned and characterized three *P. vulgaris* PIP aquaporins (GenBank ID: U97023, AY995196 and AY995195), whose best BLAST hits were, according to our nomenclature, PvPIP1;2 (U97023), PvPIP1;3 (AY995196) and PvPIP2;5 (AY995195). Under drought stress, only PvPIP1;2 was up-regulated in leaves, while all the three PIPs were up-regulated in roots under similar stress (Aroca et al. 2006).

Transcriptome analysis in common bean under drought stress using EST approaches showed contrasting expression patterns between tolerant and susceptible genotypes (Montalvo-Hernández et al. 2008; Recchia et al. 2013). Susceptible genotypes showed a marked down-regulation of aquaporins expression in response to drought, while resistant showed a stable or higher expression of these genes. In particular, under drought stress, a putative TIP aquaporin was highly expressed in root phloem tissues of a drought-resistant genotype compared to the susceptible one (Montalvo-Hernández et al. 2008). The best BLAST hit of this aquaporin, identified and cloned in a drought-tolerant (GenBank ID: DQ087217) and a susceptible (GenBank ID: DQ087218) genotype, is PvTIP2;3. Interestingly, three of the previously characterized drought-responsive aquaporins were identified as putative molecular targets of selection during common bean domestication. In particular, PvPIP2;5 was identified as putatively selected during the Mesoamerican gene pool domestication, whereas PvPIP1;3 and PvTIP2;3 were putatively selected during the Andean gene pool domestication (Schmutz et al. 2014).

Two different classes of transcription factors involved in drought response were previously characterized in wild and domesticated common beans. Two *Asr* genes, involved in an ABA-dependent drought response (Cortés et al. 2012a), and two *Dreb* genes, involved in an ABA-independent drought response (Cortés et al. 2012b). Both studies showed a higher genetic diversity and signature of selection in wild common bean in comparison to domesticated ones. Even though additional studies are needed, we suggest that PvPIP1;3, PvPIP2;5, and PvTIP2;3 could show similar patterns of diversity and selection, which could be a potential source of drought tolerance traits in cultivated common bean.

The EST database-mining analysis showed a marginal but not significant increase of PvTIP2;3 expression in response to drought stress. The only aquaporin that showed a significant increase of expression was PvTIP1;1. This result suggests a possible role for PvTIP1;1 in response to drought stress in common bean. One of the mechanisms for avoiding drought stress in common bean is an increased root depth (Beebe et al. 2013). The expression pattern of AtTIP1;1 promoter is correlated with elongating cells in

Arabidopsis roots (Ludevid et al. 1992). In addition, over-expression of *P. ginseng* PgTIP1 in transgenic *Arabidopsis* plants determined an increased tolerance to drought stress in comparison to wild-type plants (Peng et al. 2007). Thus, we propose a putative role for PvTIP1;1 aquaporin in regulating drought response in common bean.

Despite the possible importance of aquaporin in regulating drought stress response in common bean (Aroca et al. 2006; Montalvo-Hernández et al. 2008; Recchia et al. 2013), none of the drought QTLs identified by Mukeshimana et al. (2014) are located near aquaporin loci. The response variables analyzed in the above-mentioned study were mainly related to flowering time, maturity, and seed productivity, suggesting that aquaporins could have only a marginal role in these developmental processes in common bean. Early flowering and maturity are known escape mechanisms in drought tolerance, but do not represent biochemical or physiological tolerance mechanisms to which aquaporins might contribute. Nonetheless, we cannot exclude that aquaporins could be important in regulating drought stress in other bean cultivars.

In summary, the recent release of the *P. vulgaris* genome sequence enabled us to comprehensively identify and analyze the aquaporin protein family in common bean at the structural and sequence levels. We identified 41 full-length aquaporins in the common bean genome, divided in PIPs, TIPs, NIPs, SIPs, and XIPs, based on a phylogenetic analysis. Analysis of conserved structural residues showed a sub-family-specific differentiation that is conserved across different plant species. Through these residues, we inferred putative functions and substrate specificities of common bean aquaporins, based on the functional characterization of homologous proteins in other species. Analysis of expression datasets of common bean, in the forms of RNA-sequencing and EST databases, suggests that few aquaporins regulate water homeostasis and drought stress response in *P. vulgaris*. We suggest that these aquaporins (PvTIP1;1, PvPIP2;4 and PvPIP1;2) are the most physiologically important in common bean, and that research efforts should focus on the characterization of these genes first. In addition, these results could suggest a reduced functional redundancy of aquaporin genes in common bean. Tissue expression redundancies of aquaporin transcripts have been observed in other plant species previously (Nguyen et al. 2013; Reuscher et al. 2013; Zhang et al. 2013). This redundancy has been detrimental, if not misleading, for in vivo functional studies with knockdown mutants in *Arabidopsis* (Ma et al. 2004; Schüssler et al. 2008; Beebo et al. 2009). Thus, the observed reduced redundancy could facilitate physiological and functional in vivo studies aquaporins common bean aquaporins. In particular, PvTIP1;1 has the highest FPKM expression in the RNA-Sequencing datasets analyzed. It is also the only aquaporin showing significant

differential expression based on EST database mining. These results suggest an important role for PvTIP1;1 in regulating water homeostasis, plant hydraulic and drought response in this species. Nonetheless, further experiments aiming at the functional characterization of the identified aquaporins are needed for a better understanding of the molecular and physiological role of these genes in common bean.

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Ethical standards This article does not contain any studies with human participants or animals performed by any of the authors.

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